

Sign Reversal of the DC-Sweep Capacitance–Voltage Hysteresis Window in $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ Ferroelectric Transistors

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Abstract—The memory window of an HfO_2 ferroelectric transistor (FeFET) shrinks with repeated writing, and this is usually blamed on the ferroelectric wearing out. We find that the ferroelectric is not the cause. In a n-channel FeFET, the C – V hysteresis window does not just shrink to zero; it changes sign after a few thousand cycles. In a fresh device a positive write lowers the threshold voltage; after enough cycling the same write raises it. We see this reversal on five devices, at every measurement frequency and sweep amplitude, and whether the measurement is fast or slow, so it is not an artifact. Transfer measurements show the same reversal. Yet the switchable polarization does not fall over this range — it rises by 45% and stays high. The cause is charge trapped at the interface, which first cancels and then outweighs the polarization seen by the channel. Endurance here is set by this interface charge, not by the ferroelectric.

Index Terms—Ferroelectric field-effect transistor, hafnium oxide, endurance, hysteresis window, charge trapping, capacitance–voltage, interfacial layer, PUND.

I. INTRODUCTION

FERROELECTRICITY in doped HfO_2 [1], [2] made a CMOS-compatible ferroelectric field-effect transistor (FeFET) practical [3], [4]. Endurance remains the principal obstacle: the memory window typically collapses between 10^3 and 10^5 write cycles [5]–[7], while metal–ferroelectric–metal capacitors on the same material endure far longer [8]–[10].

There are two common explanations, and they point to opposite fixes. One is ferroelectric fatigue: the film loses its switchable polarization [11]. The other is charge trapping: electrons are trapped near the interfacial layer between the ferroelectric and the silicon, and this trapped charge screens the polarization so the channel no longer feels it [12]–[15]. If fatigue is the cause, the ferroelectric must be improved; if trapping is the cause, the interface must be improved. Telling the two apart therefore matters. Charge trapping and trap generation are already established in these devices [16]–[18], as is the redistribution of oxygen vacancies during cycling [19]–[21].

Two recent studies established the background for this work. In [22], ferroelectric polarization was shown to survive

after the memory window collapses; in [23], the fraction of that polarization reaching the channel was quantified and found to drop from 3.5% to 0.9% by 10^4 cycles. Both show that the polarization-to-channel coupling weakens while the polarization itself persists. The present work takes this as its starting point rather than a new result.

Three findings in the present work are not contained in those studies. First, the hysteresis window does not merely approach zero — it becomes *negative*. A coupling factor is non-negative by construction, so the regime in which interfacial charge *over-compensates* rather than merely screens the polarization field cannot be expressed in that framework. Beyond the crossing a positive gate sweep leaves the transistor harder to turn on, the opposite of its pristine behaviour. Second, the reversal is established on five independent devices, across three decades of measurement frequency, four sweep amplitudes and a ten-fold change in integration time. Third, it is resolved from the gate-stack C – V alone and independently confirmed by transfer characteristics.

II. DEVICE AND MEASUREMENT

A. Gate Stack

The devices are p-Si channel FeFETs with a $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(10\text{ nm})/\text{SiO}_2(0.85\text{ nm})$ gate stack [Fig. 1(a)]. The chemical SiO_2 interfacial layer (IL) is unavoidable on a silicon channel and is the region of interest. The insulator stack forms a series capacitive divider,

$$\frac{V_{\text{HZO}}}{V_{\text{ins}}} = \frac{C_{\text{IL}}}{C_{\text{IL}} + C_{\text{HZO}}}, \quad (1)$$

Taking $\epsilon_{\text{HZO}} = 30$ for the orthorhombic ferroelectric phase [24] and $\epsilon_{\text{IL}} = 3.9$ for SiO_2 , gives $C_{\text{HZO}} = 2.66\text{ }\mu\text{F cm}^{-2}$, $C_{\text{IL}} = 4.06\text{ }\mu\text{F cm}^{-2}$ so the ratio in (1) is 0.60. This is the largest fraction the ferroelectric can see, because the semiconductor capacitance, which is also in series, is left out; adding it would only lower the fraction. With the measured coercive voltage $V_c = 2.08\text{ V}$, this fraction corresponds to a coercive field of about 1.25 MV cm^{-1} in the film.

B. Devices Measured

Five devices from the same wafer are reported (device-1 to device-5), processed together and cycled under identical conditions, so any device-to-device difference reflects normal process variation across the wafer rather than a change in structure. Device-1 is the device from the first submission;

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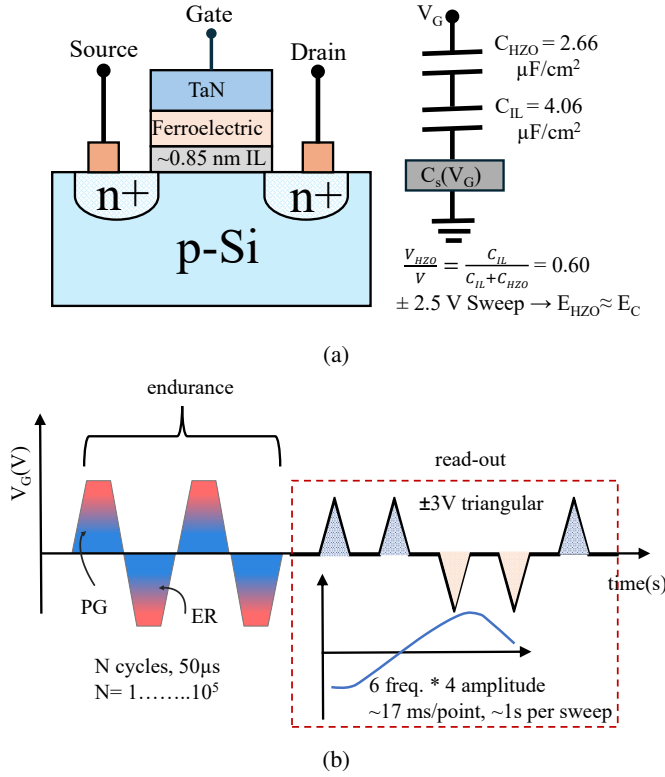


Fig. 1: Device and measurement. (a) The gate stack and its series-capacitor divider [(1)]: at most 60% of the gate voltage reaches the ferroelectric, giving $E_c \approx 1.25 \text{ MV cm}^{-1}$ from the measured $V_c = 2.08 \text{ V}$. (b) Each checkpoint: cycle with $\pm 5 \text{ V}$, $50 \mu\text{s}$ pulses, then measure. The C - V sweep is $\sim 10\,000\times$ slower than the write pulse, so it reads DC-sweep hysteresis, not a pulse-written state.

devices 2 to 5 are new. Polarization (PUND) and transfer characteristics were each measured on one device. The common-reference $2P_r$ is from a single device, consistent with the polarization behaviour already reported for this stack [22], [23]. It only confirms that $2P_r$ stays at or above its fresh value across the range where the window collapses. Reproducibility of the sign reversal is carried by the C - V data, which were taken on all five devices.

C. Polarization Measurement

Switching-current and P - V measurements used triangular waveforms with a half-period of $500 \mu\text{s}$. Polarization was evaluated by positive-up-negative-down (PUND) measurement, with

$$Q_{\text{sw}}^+ = \int [I_P - I_U] dt, \quad Q_{\text{sw}}^- = \int [I_N - I_D] dt, \quad (2)$$

where P , N denote switching and U , D nonswitching pulses, and

$$2P_r = \frac{|Q_{\text{sw}}^+| + |Q_{\text{sw}}^-|}{2A}. \quad (3)$$

Coercive voltages were taken from the P - V zero crossings, giving

$$V_{\text{imp}} = \frac{V_c^+ + V_c^-}{2}, \quad E_c = \frac{|V_c^+| + |V_c^-|}{2d}. \quad (4)$$

For the pristine device $V_c = 2.08 \text{ V}$, which with (1) corresponds to $E_c \approx 1.25 \text{ MV cm}^{-1}$, within the range reported for $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ [25], [26].

D. Electrical Measurement

Endurance cycling used $\pm 5 \text{ V}$, $50 \mu\text{s}$ pulses. At each checkpoint we measured the polarization by PUND and the capacitance by C - V on a Keysight B1500A (parallel C_p - R_p mode, 20 mV AC). Devices 4 and 5 were measured at seven frequencies (1 kHz to 1 MHz) and four sweep amplitudes ($\pm 1.5 \text{ V}$ to $\pm 3 \text{ V}$); devices 2 and 3 at 100 kHz and $\pm 3 \text{ V}$, and repeated at two integration times. Polarization was obtained from $2P_r = (|Q_{\text{sw}}^+| + |Q_{\text{sw}}^-|)/2A$, with the switching charges from the standard PUND subtraction, and the coercive and imprint voltages from the zero crossings of the P - V loop.

One feature of the C - V measurement must be stated because it sets what the window means. The sweep starts at the negative rail and spends about one power-line cycle on each point, so a full sweep takes roughly one second — four orders of magnitude longer than the write pulse. A pulse-written state cannot survive that long during the sweep, and indeed the C - V curves taken after a program pulse and after an erase pulse lie on top of each other. What we report is therefore the *DC-sweep hysteresis window*, the gap between the forward and reverse branches at the mid-point of the flank,

$$\Delta V_{\text{hyst}} = V_{\text{fwd}} - V_{\text{rev}} \Big|_{C=C_{\min} + \frac{1}{2}(C_{\max} - C_{\min})}. \quad (5)$$

This is not the same as the pulse-written memory window; the transistor version of that is measured directly in Section III-C. We fix the sign so that $\Delta V_{\text{hyst}} > 0$ is the normal ferroelectric loop (a positive sweep lowers V_{th}) and $\Delta V_{\text{hyst}} < 0$ is the trapping loop (a positive sweep raises V_{th}).

Transfer characteristics used an SPGU to apply the write pulse and an SMU to read, with $V_D = 0.2 \text{ V}$ and V_{th} taken at $I_D = 1 \mu\text{A}$. Switching the gate between the pulse and the read through the switching matrix takes about 60 s , which is the read delay. Relaxation after a write was measured separately. A program pulse was applied to the gate; the gate was then held at the read level and the drain current was sampled from 2 ms to 1000 s , with $V_D = 0.2 \text{ V}$ and the source and bulk grounded.

E. Corrections, Screening, and Analysis

To check whether the hysteresis is affected by charge leaking during the ADC integration, the cycling series was repeated on devices 2 and 3 with the integration time increased from 1 to 10 power-line cycles (16.7 ms to 167 ms per point), all else unchanged. A series resistance was extracted from the Nicollian-Brews strong-accumulation expression [27], [28]

$$R_s = \frac{G_{\text{ma}}}{G_{\text{ma}}^2 + \omega^2 C_{\text{ma}}^2}, \quad (6)$$

and capacitances corrected using

$$C_c = \frac{(G^2 + \omega^2 C^2)C}{a^2 + \omega^2 C^2}, \quad a = G - (G^2 + \omega^2 C^2)R_s. \quad (7)$$

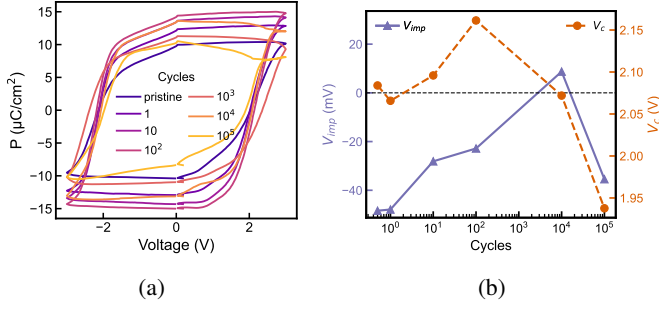


Fig. 2: Ferroelectric response versus cycling. (a) P-V loops at each cycle count; the loop opens with cycling and does not shrink. (b) Imprint voltage V_{imp} (left) and coercive voltage V_c (right): V_c holds near 2 V, so the switching field is unchanged, while V_{imp} drifts steadily from -48 mV toward positive values, the signature of a growing interface charge on the channel side.

Sweeps with a median loss tangent $D = 1/\omega R_p C_p$ above the stated screen were excluded from the window statistics.

An unscreened remanent polarization would produce a window $2P_r/C_{IL}$, so we define the window efficiency

$$\eta = \frac{\Delta V_{hyst}}{2P_r/C_{IL}}, \quad (8)$$

Where a window deficit is attributed to interfacial charge, an equivalent areal density follows from

$$\Delta N_{eq} = \frac{C_{IL} \Delta V}{q} \approx \Delta V \times 2.5 \times 10^{13} \text{ cm}^{-2} \text{ V}^{-1}. \quad (9)$$

Equation (9) assumes a sheet charge at the interfacial layer and an unchanged C_{IL} . It is an equivalent density inferred from a voltage shift, not a directly extracted interface-trap density, and is denoted N_{eq} for that reason.

III. RESULTS

A. The Ferroelectric Wakes Up, Then Fatigues

Fig. 2 shows how the ferroelectric responds to cycling. In the first hundred cycles the film wakes up, a behaviour reported for this material and attributed to oxygen-vacancy redistribution and domain-wall de-pinning [21], [29], [30]: the P-V loop opens up and $2P_r$ rises from $20.2 \mu\text{C cm}^{-2}$ to $29.2 \mu\text{C cm}^{-2}$ by 10^2 cycles, a 45% increase. The coercive voltage stays near 2 V the whole time [Fig. 2(b)], so the switching field itself does not change — the film is not degrading during wake-up, it is simply switching more completely. $2P_r$ stays at or above its fresh value through 10^4 cycles.

The imprint voltage tells the rest of the story. Over the same wake-up range V_{imp} moves by 26 mV, from -48 to -23 mV, and continues to 9 mV by 10^4 cycles, crossing zero between 10^3 and 10^4 [Fig. 2(b)]. This drift is the mark of negative charge collecting on the channel side of the film [26], and it is already underway while the polarization is still rising. The trapping therefore does not wait for the wake-up to finish; the two run together from early in the cycling.

The ferroelectric itself only fatigues much later. At 10^5 cycles $2P_r$ falls to $18.6 \mu\text{C cm}^{-2}$, and the coercive voltage

drops from 2.08 V to 1.94 V — not the signature of domain-wall pinning, but consistent with a leakier film in which trap-assisted conduction contributes [31]. So the ferroelectric does degrade, but only about two decades after the hysteresis window has already collapsed.

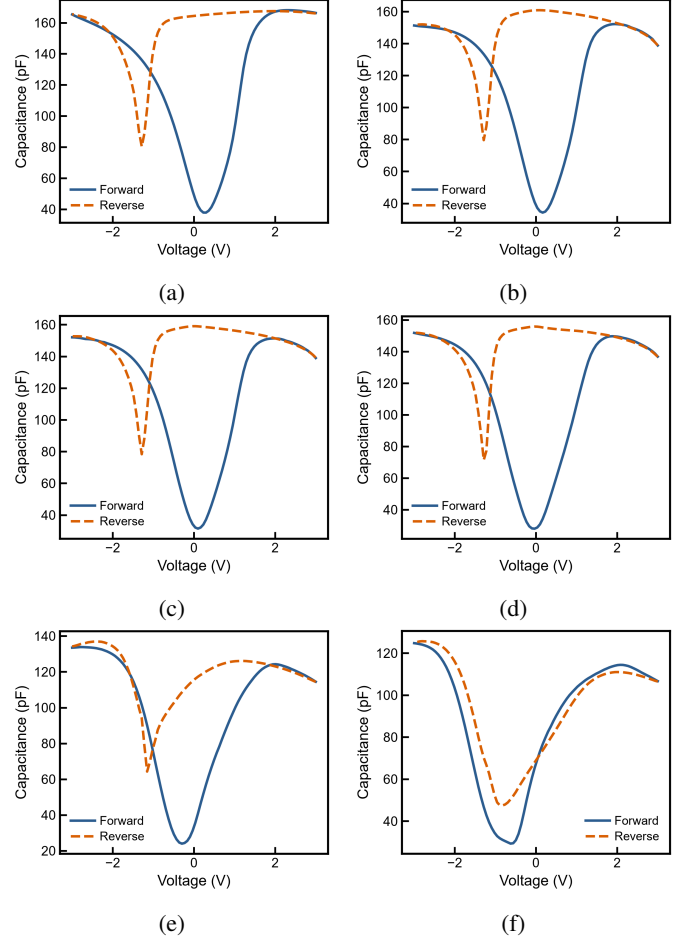


Fig. 3: C - V evolution of device-7 at 500 kHz, ± 3 V. Solid: forward sweep; dashed: reverse. The separation narrows monotonically, closes near 10^3 cycles, and reopens at 10^4 with the branches exchanged.

B. The Window Collapses and Reverses Sign

Fig. 3 shows the C - V evolution: the separation between the forward and reverse branches narrows with cycling, closes near 10^3 cycles, and reopens at 10^4 with the two branches exchanged.

Fig. 4(a) shows ΔV_{hyst} for five independent devices. All start with a counter-clockwise window when fresh and end with a clockwise window by 10^4 cycles; the window crosses zero at 503, 871, 1117, 2136 and 2422 cycles [Fig. 4(b)], a spread of a factor of five. At 10^4 the window lies between -0.15 and -0.28 V on all five (Table I). Device-1 has a smaller fresh window (0.363 V) than the other four (0.79–0.82 V), a die-to-die variation, but the sign reversal is common to all.

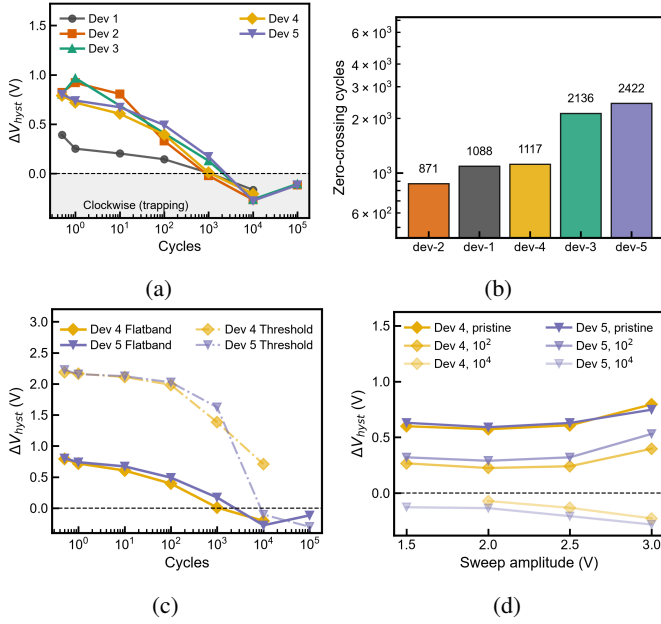


Fig. 4: The window collapses and reverses sign. (a) ΔV_{hyst} versus cycling for five independent devices; shaded region marks clockwise loops. Device-1 (grey) has a smaller pristine window but the same trajectory. (b) Interpolated zero crossings. (c) Flatband and threshold flanks, devices 4 and 5, both inverting on the same schedule. (d) Window versus sweep amplitude at 1 MHz; the reversal is present at every amplitude.

The window can be read at two points on the C - V curve: near threshold, on the inversion side that sets V_{th} , and near flatband, on the accumulation side. Both reverse on the same schedule [Fig. 4(c)], so the reversal is not an artifact of where the window is measured. The threshold-side window is consistently the larger of the two, which means the loop stretches as it cycles rather than shifting sideways as a whole; a fixed sheet of trapped charge would shift both points by the same amount and could not do this. The reversal also appears at every sweep amplitude [Fig. 4(d)]. Because the 50%-of-range definition is not directly comparable between different amplitudes, we take from this only that the sign change is always present, not a trend in its size.

C. The Reversal Is a Device Property, Not an Artifact

Four checks establish that the sign reversal is a real device behaviour and not a measurement or gate-stack effect. First, ΔV_{hyst} is suppressed at low frequency and saturates above ~ 100 kHz, following

$$\Delta V_{\text{hyst}}(f) = \Delta V_{\infty} - \frac{\delta V}{1 + (f/f_c)^2}, \quad (10)$$

with a corner frequency $f_c = 25.6$ kHz (device-4) and 16.8 kHz (device-5) when fresh [Fig. 5(a)]. Across the fresh and lightly-cycled checkpoints, where the fit is well constrained, f_c stays in the range 17–26 kHz on all the devices, and the reversal is present at every one of the seven measurement frequencies.

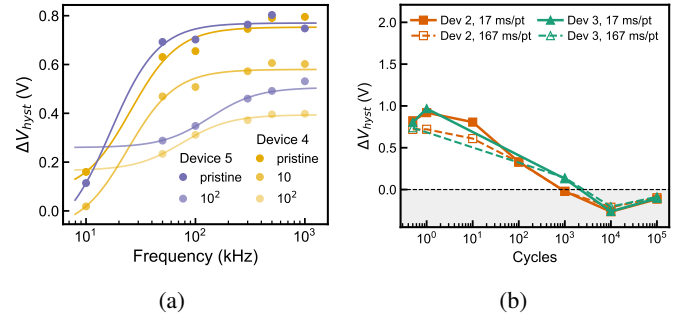


Fig. 5: The reversal is not a measurement artifact. (a) Inverted-Debye dispersion, (10), devices 4 and 5; solid curves are fits, with $f_c \approx 17$ –26 kHz in the pristine and lightly-cycled state, consistent with device-1. (b) PLC 1 versus PLC 10 (16.7 versus 167 ms per point); the sign is unchanged at every measurement point through 10^5 , and the magnitude moves toward zero at longer integration, opposite to the dependence expected from a leakage artifact.

Second, making each measurement point ten times slower changes the size of ΔV_{hyst} by only 15–22% and never flips its sign, at any measured cycle count on either device [Fig. 5(b)]. This rules out leakage during the measurement as the cause of the negative window. If leakage were responsible, a slower measurement — which gives charge more time to leak — would make the window *more* negative, not less. We see the opposite: the window shrinks toward zero at the slower setting, so leakage cannot be the cause.

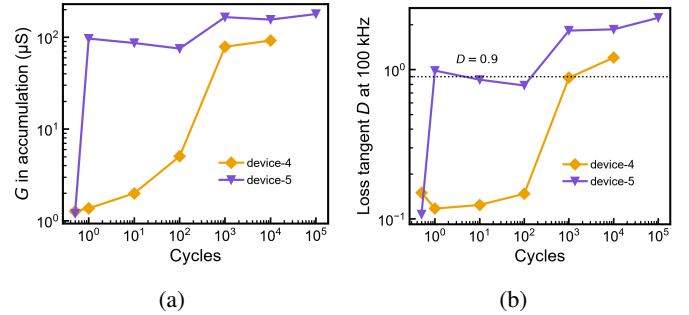


Fig. 6: The interface degrades with cycling (devices 4 and 5). (a) Accumulation conductance rises about 100 $\times$, showing leakage growing as defects accumulate. (b) Loss tangent D at 100 kHz grows past $D = 0.9$ (dotted), where leakage becomes comparable to the capacitive response. Both are read directly from the measurement and need no series-resistance correction.

Together, these checks show that the reversal is a property of the device, not of the measurement. Its cause is a physical change at the interface, which the gate-stack data make clear [Fig. 6]. As the device is cycled, the accumulation conductance rises by about 100 $\times$ [Fig. 6(a)], and the loss tangent grows past $D = 0.9$ [Fig. 6(b)], where leakage becomes comparable to the capacitive response. Unlike the capacitance, the conductance and loss tangent are read directly from the measurement and need no series-resistance correction, so they stay reliable across the full cycling range. This growing interface loss is the

expected signature of defects accumulating at the interfacial layer, and it develops over the same cycling range in which the window reverses — independent evidence for the interface charge that drives the reversal.

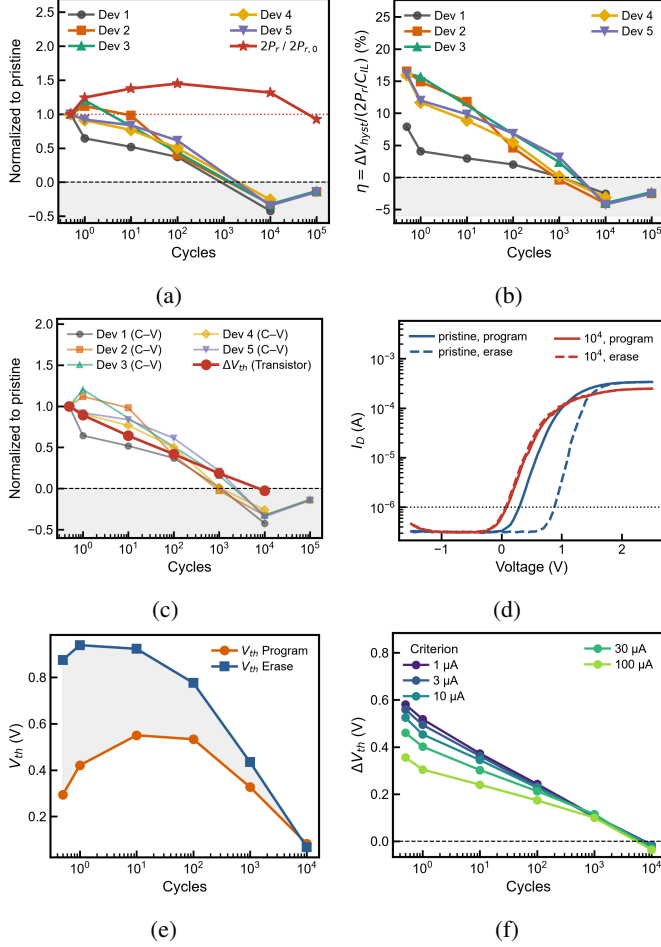


Fig. 7: Polarization and window are decoupled. (a) ΔV_{hyst} normalised to each device's own pristine value, five devices, with $2P_r/2P_{r,0}$ from the common reference measurement. (b) Window efficiency (8) against that reference. (c) Normalised ΔV_{th} (transistor, single device, 60 s read delay) plotted against the normalised ΔV_{hyst} of all five devices.

D. Transistor-Level Confirmation

Fig. 8 shows transfer characteristics measured after program and erase at each checkpoint. Defining $\Delta V_{\text{th}} = V_{\text{th}}(\text{erase}) - V_{\text{th}}(\text{program})$ so that the sign convention matches ΔV_{hyst} , the transistor window falls from 0.581 V pristine to 0.243 V at 10^2 , 0.108 V at 10^3 and -0.015 V at 10^4 . The reversal is independent of the extraction criterion, being present at 1, 3, 10, 30 and 100 μA [Fig. 8(c)]. Both threshold states converge and cross, falling from 0.29 and 0.87 V pristine to 0.083 and 0.067 V at 10^4 .

The assumption stated in Section II-D is therefore supported by direct measurement: the capacitance and transistor windows collapse on the same schedule and change sign at the same cycle count [Fig. 7(c)]. The two are measured on very different timescales — 60 s for the transistor read against ~ 17 ms

per point for $C-V$ — so their agreement indicates that the responsible charge is stable over at least that range. The off-state current floor of $\sim 3 \times 10^{-7}$ A in these devices prevents extraction of the subthreshold swing.

A post-write relaxation transient is shown in Fig. 9 At the pristine state the drain current decays by 3.5% over five decades, well described by a stretched exponential with $\tau = 0.10$ s and $\beta = 0.84$; the fit is well constrained only at this checkpoint. The relaxation amplitude is essentially unchanged at 10^3 cycles (3.8%), indicating that the pre-existing population of fast interface traps — filled during the write and emptied over ~ 0.1 s — is not modified between the pristine state and 10^3 cycles. At 10^4 cycles the transient is no longer resolvable. We note that its amplitude scales with dI_D/dV_{th} at the read point as well as with the relaxing charge, so the loss of a resolvable transient is consistent with, but does not by itself establish, a change in that population. The monotonic reversal of ΔV_{hyst} and ΔV_{th} over the same range is therefore attributed to a cycling-generated trap density that grows with N , distinct from the fixed fast population probed here, in line with the trap generation reported for this system by charge pumping [17] and low-frequency noise [18]. A trap time constant is not extracted beyond the pristine checkpoint.

E. Polarization and Window Are Decoupled

Fig. 7 places the measurements together. Normalising each device's window to its own pristine value [Fig. 7(a)], all five fall to between -0.26 and -0.42 of their initial magnitude at 10^4 cycles, while the reference $2P_r$ rises to 1.45 of its pristine value at 10^2 and remains above unity at 10^4 .

The window efficiency of (8), evaluated against the common reference polarization, falls from 15.9–16.5% pristine to -3.2 to -4.2% at 10^4 cycles for devices 2, 3, 6 and 7 [Fig. 7(b), Table I; device-1 follows the same trajectory from a lower starting value. The agreement among four independent devices at both ends of the range is closer than the device-to-device spread in the pristine window itself.

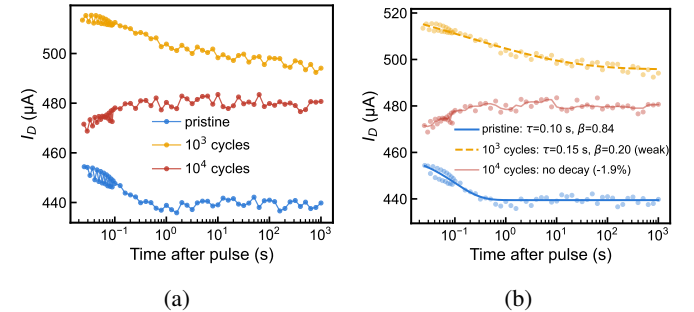


Fig. 8: Relaxation after a 5 V, 50 μs program pulse, measured by fixed-gate drain-current sampling. (a) Raw transients at three checkpoints. (b) Stretched-exponential fits to the running median; the pristine trace gives $\tau = 0.10$ s, $\beta = 0.84$, while the 10^3 fit is unconstrained (dashed) because τ falls below the first sample and the 10^4 trace shows no decay. The relaxation amplitude is 3.5–3.8% and is reported as an observation only.

IV. DISCUSSION

A. Origin of the Sign Reversal

In a DC sweep of an MFIS stack the two hysteresis mechanisms rotate the loop in opposite senses. Ferroelectric switching produces a counter-clockwise loop: the forward excursion polarizes the film toward the channel, so the device turns on at a lower gate voltage on the return branch and $\Delta V_{\text{hyst}} > 0$. Charge trapping produces a clockwise loop: the same forward excursion injects electrons into states at and near the interfacial layer, raising V_{th} on the return branch, so $\Delta V_{\text{hyst}} < 0$. The measured quantity is therefore a difference between two positive magnitudes,

$$\Delta V_{\text{hyst}} = \Delta V_{\text{FE}} - \Delta V_{\text{trap}}, \quad (11)$$

and nothing requires the second to remain the smaller of the two.

Cycling moves the two terms in opposite directions. ΔV_{FE} is bounded above by $2P_r/C_{\text{IL}}$ and can only decrease as compensating charge accumulates between the ferroelectric and the channel; the PUND data show that this decrease is not caused by loss of polarization, since $2P_r$ is larger at 10^4 cycles than pristine. ΔV_{trap} scales with the interfacial charge exchanged per sweep, which grows with the density of available states; trap generation during cycling is well documented in this system [12], [13], [17]. A bounded decaying term subtracted from a growing one crosses zero, and the cycle count at which it does so depends on the initial window and the initial trap density — consistent with the factor-of-five spread observed across the five devices.

TABLE II
Reported Endurance of HfO₂-Based Ferroelectric Devices

Ref.	Structure	Ferroelectric / IL	Endurance	Window extracted	Sign reversal
	MFMC capacitor	Y:Hf _{0.5} Zr _{0.5} O ₂	-	$2P_r$	no
	p+ FE-MOSCAP	HfZrO ₂ / SiO ₂	window zero at ~ 500	C-V	no
	22 nm FDSOI FeFET	HZO / SiO ₂	$\sim 2 \times 10^4$	ΔV_{th}	no
	FeFET	HZO / SiO ₂	$\sim 10^4$	ΔV_{th}	no
	BEOL FeFET	HZO / IWO	$\sim 10^{11}$	ΔV_{th}	no
	This FeFET	HZO / SiO₂	beyond 2×10^5, negative	C-V, $2P_r$ and ΔV_{th}	yes

Endurance in every prior report is quoted as the cycle count at which the memory window closes or falls below a threshold. In each case the window is treated as a non-negative quantity decaying toward zero. The present work reports the window passing *through* zero and becoming negative, on five devices, with the same reversal confirmed in the transistor threshold window. Rows marked [X] are to be completed with the primary citation; entries traced through a secondary source are noted and the primary paper should be cited.

B. Consistency of the Observations

Four features of the data support this reading. The flank asymmetry indicates bias-dependent interface states rather than a fixed charge sheet, which would displace both flanks

equally. The imprint drift is in the direction expected from a net negative sheet on the channel side [26]. The frequency dispersion of (10) shows that part of the compensating charge responds on a microsecond timescale: charge given time to respond screens the window, charge that cannot follow the AC signal does not. The accumulation conductance increases with cycling, consistent with trap-assisted transport through a defect-rich interfacial region.

From (9), the 0.51 V swing in ΔV_{hyst} on device-1 corresponds to $\Delta N_{\text{eq}} \approx 1.3 \times 10^{13} \text{ cm}^{-2}$, and the 1.08 V swing on device-5 to $\approx 2.7 \times 10^{13} \text{ cm}^{-2}$. Both lie within the range reported for degraded HfO₂/SiO₂/Si interfaces [12], [14], [17].

C. Limitations

The measurements do not determine where the compensating charge resides. They do not distinguish charge in bulk HZO, at the HZO/SiO₂ interface, within the SiO₂, or at the SiO₂/Si interface, and we describe it only as effective charge within the channel-side gate stack. The mechanism is presented as consistent with charge trapping rather than as an identification of the responsible states; transmission electron microscopy with electron energy-loss spectroscopy, or charge pumping across the cycling range, would be required to localize and quantify it directly.

The population probed by the AC dispersion, with $\tau \approx 6 \mu\text{s}$, is not the population responsible for the DC sign reversal, which is established over the $\sim 17 \text{ ms}$ dwell of the sweep; these differ by more than three decades in time constant and no common origin is claimed. We note additionally that δV in (10) does not grow with cycling, so the AC-active population is not itself the one that inverts the loop.

Finally, the five devices are from a single die, and the zero-crossing cycle count varies by a factor of five among them. Polarization and transfer characteristics were each measured on a single device (Section II-B), so the decoupling argument combines a five-device window measurement with single-device polarization and transistor references. The value of η depends on the convention chosen for the ideal window in (8); alternative denominators shift the absolute values but not the sign change or the trend.

V. CONCLUSION

The DC-sweep C-V hysteresis window of an HfO₂ FeFET does not merely close with cycling: it reverses sign. The reversal is reproduced on five independent devices with zero crossings between 5×10^2 and 2×10^3 cycles, survives a ten-fold increase in measurement integration time, is present at seven measurement frequencies and four sweep amplitudes, and is confirmed by transfer characteristics in which ΔV_{th} changes sign at the same cycle count. Over the same range the remanent polarization stands above its pristine value, and where the lumped C_p - R_p model is valid the gate-stack capacitance is constant to better than 1% while 64% of the collapse occurs. Loss of switchable polarization alone therefore cannot account for the early collapse of the hysteresis window in these devices. The observations are consistent with interfacial charge that first compensates and then over-compensates the polarization field at the channel.

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